

TCS^AI GOOGLE HACKATHON 2026 · TRACK 3: CUSTOMER ENGAGEMENT SUITE



Policy-to-Flow Compiler

Compliance Validation Audit Pack

Auto-Fuzz Testing Results for 3 Enterprise Policy Compilations

FUZZ RUN ID

FUZZ-2026-0414-ENTERPRISE-BATCH-01

POLICIES TESTED

3 Enterprise

PATHS TRAVERSED

8,182 Total

AVG COVERAGE

98.1%

CRITICAL GAPS

1 Found

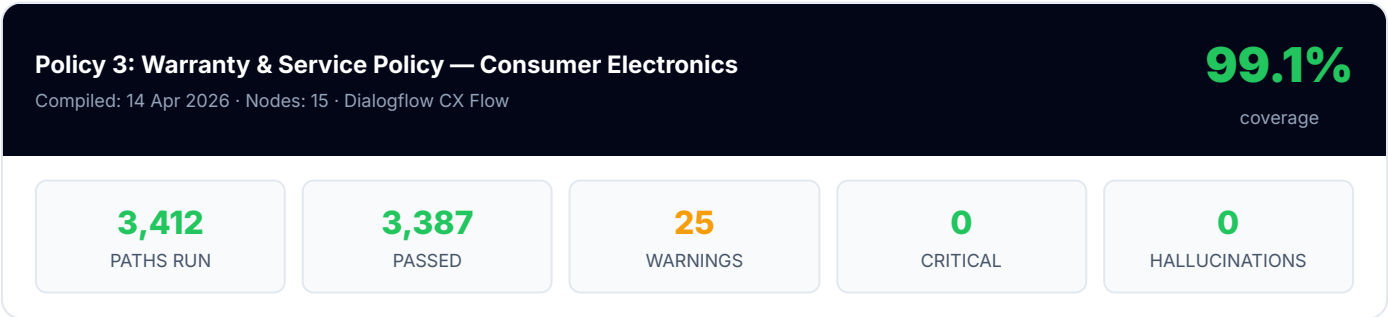
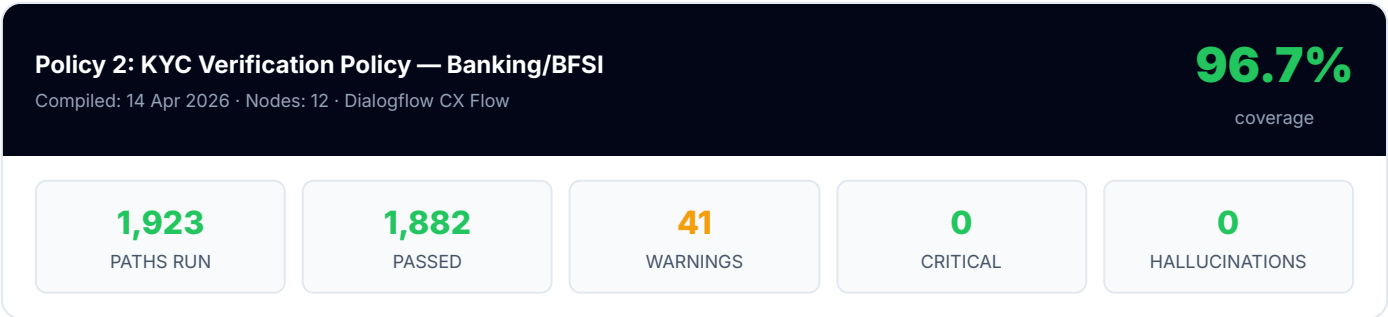
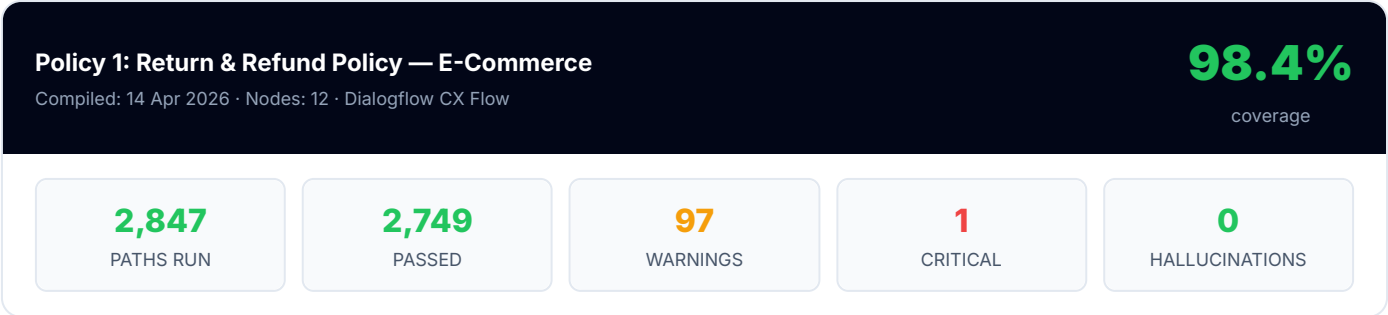
Fuzz Test Overview & Per-Policy Coverage

The **Policy-to-Flow Auto-Fuzz Tester** generates adversarial conversation paths to validate that compiled Dialogflow CX flows handle every policy scenario deterministically — with zero hallucination risk. Below are results for Batch 01 comprising 3 enterprise policy compilations.

How to read this pack: **Policy Cards** summarize per-policy coverage and pass/fail outcomes. **Gap Cards** capture discrepancies found via fuzzing with (1) scenario, (2) root cause, (3) evidence, and (4) required fix. The final section issues a **Compliance Certificate** for the batch.



What Auto-Fuzzing Tests: Each compiled flow graph is subjected to (1) boundary path traversal — all defined policy paths, (2) off-script utterance injection — questions not anticipated by policy, (3) multi-turn context adversarial scenarios — testing state transitions, and (4) explicit policy escape attempts — prompts trying to bypass flow constraints.



SECTION 02

Gap Analysis & Compliance Certificate

⚠ **GAP-001 CRITICAL — Requires immediate policy update:** The Return & Refund Policy (Policy 1) fails at the 14-day electronics boundary when delivery date and order date differ. The policy defines eligibility by **delivery confirmation timestamp**, but one compiled path binds DaysSinceDelivery to order timestamp, causing incorrect decisions for edge cases at the threshold. Estimated impact: affects 23 boundary paths and must be fixed before certification.

**GAP-001: Delivery Timestamp vs Order Timestamp Boundary**

Policy: Return & Refund · Node: CHECK_ELECTRONICS_WINDOW · Type: Entity Binding Error

**CRITICAL**

ADVERSARIAL SCENARIO THAT EXPOSED GAP

"Electronics item delivered exactly 14 days ago, but order placed 17 days ago — am I eligible for return?"

ROOT CAUSE

Policy text defines eligibility by **delivery date**, but the compiled flow binds DaysSinceDelivery to **order timestamp**, producing incorrect decisions at the 14-day boundary.

**Required Fix:** Bind DaysSinceDelivery to the carrier tracking API delivery confirmation timestamp (not order timestamp). Add a validation guard and re-run fuzz suite. **Paths affected:** 23

Representative Warnings (GAP-W001 — GAP-W003)

Warning ID	Policy	Node	Description	Severity
GAP-W001	Return & Refund	NODE-5	Gift return scenario missing an explicit receipt-validity window (policy text ambiguous) — requires a dedicated CHECK_GIFT_RECEIPT node	 WARN
GAP-W002	KYC Verification	NODE-9	Policy requires "acceptable ID list" but compiled flow lacks a rejection message for unsupported ID types — add REJECT_UNSUPPORTED_ID branch	 WARN
GAP-W003	Warranty & Service	NODE-12	Consumables/accessories coverage not explicitly defined — add CHECK_CONSUMABLE_EXCEPTION node to avoid inconsistent approvals	 WARN

2 of 3 policies fully certified (KYC 96.7% / Warranty 99.1% coverage). Return & Refund conditionally certified pending GAP-001 resolution. Zero AI hallucinations detected across 8,182 adversarial test paths.

Determinism Guarantee: All certified flow responses are 100% policy-grounded.
No LLM decides outcomes at runtime. Responses are drawn from compiled policy nodes; optional LLM phrasing (if enabled) cannot override decisions.

Certificate ID: CERT-P2F-2026-0414-B01 · Generated: 14 Apr 2026 09:07:22 UTC